

## GE-Problem Set 5, ECN 714

**Exercise 1** Suppose there are  $I$  consumers in an exchange economy with  $L$  goods. Each consumer has a utility function parametrized by an exogenous parameter  $r_i \in [0, 1]$ . (For example, the parameter could indicate an elasticity of substitution.) Thus for each consumer  $i$ ,  $U_i : \mathbb{R}_+^L \times [0, 1] \rightarrow \mathbb{R}$ . Define  $r \in [0, 1]^I$  to be the vector of parameters for all consumers. In addition, each consumer has an initial endowment vector  $\omega_i > 0$  (i.e.  $\omega_{\ell i} \geq 0 \ \forall \ell$ ,  $\omega_{ki} > 0$  for some good  $k$ ).

Let each consumer's demand be a single-valued, continuous function on  $(p, r_i) \in \Delta \times [0, 1]$ . Let  $\phi(r)$  denote the exchange economy corresponding to the parameter vector  $r \in [0, 1]^I$ .

- Is the information provided on preferences and demand functions consistent with strongly monotone preferences?
- Show that the correspondence

$$E(r) = \{(x, p) : (x, p) \text{ is a Walrasian equilibrium in } \phi(r)\}$$

is upper hemi-continuous. (Recall that if  $X \subset \mathbb{R}^M$ ,  $Y \subset \mathbb{R}^N$  and  $Y$  is compact, then the correspondence  $f : X \rightarrow 2^Y$  is uhc if it has a closed graph.)

- Suppose that for each  $r \in [0, 1]^I$  there is a Walrasian equilibrium in the economy  $\phi(r)$ . Show that there is a Walrasian equilibrium when  $r = (1, 1, \dots, 1)$ .

**Exercise 2** Let  $A \subset \mathbb{R}^L$  be a non-empty, open set. Prove that  $A$  does not have Lebesgue measure zero.

**Exercise 3** Consider an exchange economy with two goods and  $I$  identical consumers, each with utility function  $U(x_1, x_2) = x_1^2 + x_2^2$  and initial endowment  $\omega = (1, 1)$ .

- If  $I = 2$ , what are the Walrasian Equilibria?
- If  $I > 2$ , what are the Walrasian Equilibria?
- If  $I = 2$  but the endowments are  $w_i = (1/2, 1/2)$  and  $w_{i'} = (3/2, 3/2)$ , what are the Walrasian Equilibria?